

Attack Types

Lecture 4



Outline

It includes:

Cyber attacks:

- Malware.
- Brute Force Attack
- Cross-site scripting (XSS)
- Denial-of-service (DoS) attack
- Distributed Denial of Services (DDoS)
- Man-in-the-middle attack
- Social engineering
- Phishing
- Spear phishing
- Spoofing
- language(SQL)injection
- Zero-day exploit
- Botnets



Malwares and attack Types

❖ **Malware**, also called malicious code, is software designed to:

- gain access to targeted computer systems,
- steal information or
- disrupt computer operations.

❖ **Malware and malicious files inside a computer system can:**

- Deny access to the critical components of the network
- Obtain information by retrieving data from the hard drive
- Disrupt the system or even render it inoperable



Introduction

- The **malware** is a computer software program maliciously developed to install on the computers without any consent of the users.
- The **main objective** of malware programs is to either establish the access to the targeted computers without any permission of the user or to install the program to create annoyance for the users.
- The annoyance can include pranks, data theft, data damage, computer malfunction, service blockage, and spying on the users.
- **In most of the cases**, the malwares are installed on the targeted computers to achieve different types of **goals**, which may include getting financial favor, knowing the business strategies, sabotaging the governmental system, and many others.



Malwares

The main types of malware software programs are explained below:

- **Viruses**
- **Trojan Horse**
- **Rootkit**
- **Worms**
- **Adware**
- **Scareware**
- **Browser Hijacker**



1-Viruses

❖ There are several types of malware, the most important are as follows:

➤ common types of malware include:

- **Viruses:-** A computer virus is a piece of code that can replicate itself and spread from one computer to another. It requires intervention or execution to replicate and/or cause damage.
- A **computer virus** is a malicious computer program, which is designed to alter the computer functions, slow down the computer performance, and to damage the valuable files on the computer drive.
- The **virus** are **executable files** when run on any machine by **mistake** or **through any trick**, it will activate and will start altering the computer configurations and the process, which are used for the smooth operation of the computers.
- The **virus programs** have the capability to copy into multiple files and thus, overwhelm the computer processes and data storage.



1- Viruses

- The viruses are just programs in the **form of executable files**. They don't activate without running those executable files on your computers.
- If you got a virus program on your computer, but it has not been executed on your computer, your computer is not infected as yet.
- As soon as you execute that file by either clicking or running some legitimate programs on which, the virus files are attached, the virus becomes active, and it will start doing its designed job.



1-Viruses

We can save our computers from the attack of nasty viruses by taking the following measures:

- Don't use your credit cards on the websites that don't comply with the Payment Card Industry Data Security Standard (PCI-DSS) guidelines for secure transactions.
- Never insert flash cards or other storage devices to your computer without scanning and knowing about the type of data.
- Never open emails and their attachments from unknown people.

By following the above mentioned rules of thumb, you can save your computer from the effect of computer viruses.

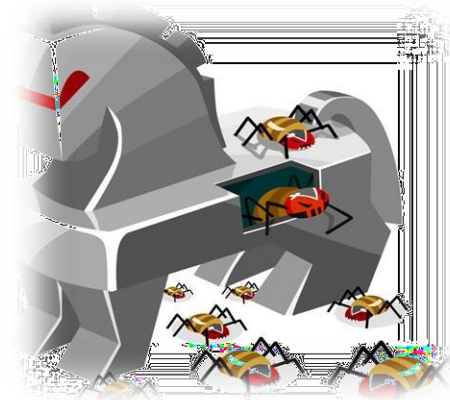


2-Trojan Horses

❖ There are several types of malware, the most important are as follows:

➤ common types of malware include:

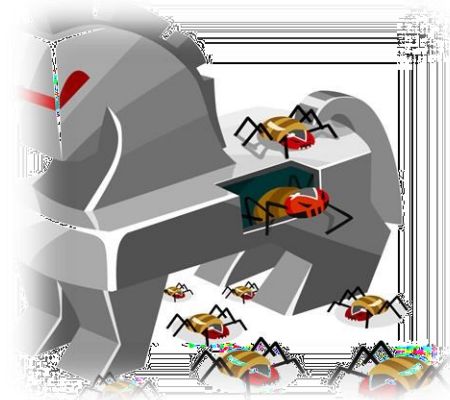
- **Trojan horses:-** A further category of malware is the Trojan horse, which is a piece of malware that gains access to a targeted system by hiding within a genuine application. Trojan horses are often broken down into categories reflecting their purposes.
 - **Trojan horse** is also a malicious computer program that may look very meek and harmless, but it can pave the way for a bigger attack on your computer and valuable data.
- It **collects the information** about the user behavior, credentials and other activities on the computer silently and sends them to its command and control center from where the other malicious attacks can originate.
- **Hackers use Trojan** to open the backdoor on your computer to access your computer and establish control on your machine.
- **The concept** of the Trojan horse in the modern computer terminology resembles 100% with the story of Trojan horse in the Greek mythology.



2- Trojan Horses

◦ Generally **recognized activities** of a Trojan horse on an infected computer include:

- Collecting data and sending to command and control center
- Copying the files and credential information
- Blocking of the data
- Altering the useful data
- Reducing the performance of computer
- Deleting some useful data files

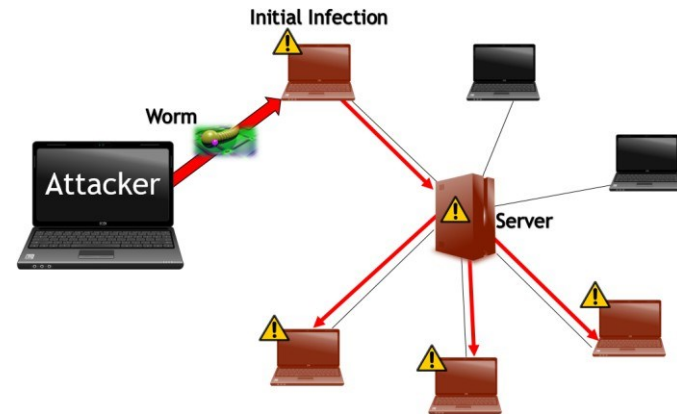


3-Network Worm

❖ There are several types of malware, the most important are as follows:

➤ common types of malware include:

- **Network Worm:-** A variant of the computer virus, which is essentially a piece of **self-replicating code** designed to spread itself across computer networks. It does not require intervention or execution to replicate.
- The **main feature** of a computer worms and viruses is that they replicate themselves in order to spread to other computers on the network or through other data transmission mediums.

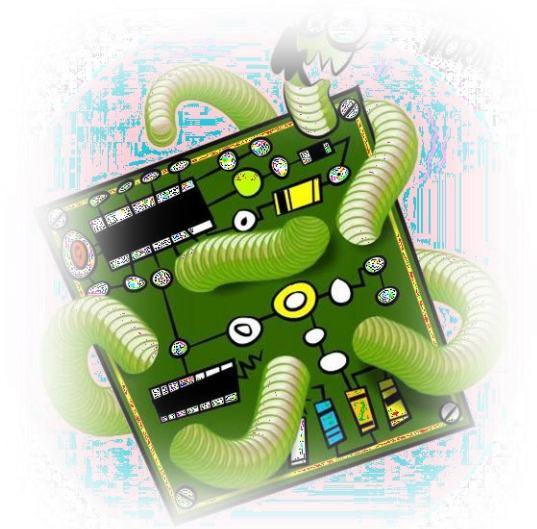


3-Network Worm

- The **first worm** code named as “**tapeworm**” to replicate on 100 different nodes for a communication was created at Xerox Palo Alto Research Center by John Shoch

The main points of a worm

- It replicates and spreads
- Exploits communication protocol vulnerabilities on networks
- Exploits operating systems on the computers
- They are harmless in nature without any payloads
- Payloads on the worms can be harmful like virus or other malicious attacks
- They infect the computers without any interaction from the user
- Bandwidth, storage space, memory can be consumed with uncontrolled replication



3- Network Worm

A worm infected computer can be recognized by certain **symptoms**. *Those symptoms are listed below:*

- Some errors related to the operating system and system files appear
- The modified files disappear or don't open
- Firewall warnings may popup in certain cases
- Strange icons or files appear on the desktop
- Computer generates unknown errors in sounds, messages or even in images
- The performance of computers slows down
- Sometimes computer may hang or freeze
- In extreme conditions system may crash frequently
- In extreme cases, you will find emails sent to your contacts without your knowledge.



4- Spyware

❖ There are several types of malware, the most important are as follows:

- **Spyware (Sniffing)**:- A class of malware that gathers information about a person or organization without the knowledge of that person or organization.
- The **main objective** of this program is to monitor online activities, computer using habits and personal interests.
- A spyware snoops the way user uses the internet.
- In most cases the spywares are used to learn the habits of the computer users so that proper and focused digital marketing campaign through emails and other online sources can be implemented.
- In the present-day modern world, the importance of **privacy** has become very critical.



4- Spyware

A few very important symptoms of your computer to have been affected by the adware include the following:

- Frequent redirects
- Huge number of spam emails
- Frequent popup windows of offers
- Bombardment of product ads in browser
- Heavy outgoing and incoming traffic
- Slowdown of internet connection



4- Spyware

Adware (Spam):- Designed to present advertisements (generally unwanted) to users

- **Adware** has become a very popular tool for digital marketing teams to draw attention of the users to a particular product or service.
- The adware code uses different ways to propagate and find the suitable targets so that **the focused audience** is figured out for a particular product.
- You might have come across some annoying links and pages while browsing the internet. When **you click on those pages or links** you are redirected to another page that promotes a particular product or service.
- In some cases, **pop-ups windows** appear with promotional content for a particular product.
- This is all done with the help of adware programs.



5- Browser Hijacker

- When you observe this situation, be sure that your computer's browser has been hijacked by the malicious program, which is altering the settings of your browser and creating a way for the malicious programs to intrude in.
- In some cases, you might have seen that some **unknown plugins** have been installed and activated on your browser. Although you have not downloaded or installed on your browser, but still they are there and active.

- **Browser Hijacker** malware is a **certain malicious code**.
- This code is used to pave the way for different kinds of cyberattacks on your computer or to the other computers on your network.
- Browser hijacker normally downloaded on your computer via some free software applications.
- The **main objective of browser hijacker** software is to force the users to visit certain websites for improving the volume of traffic on that particular website.
- When the traffic of a website is improved, the website gets higher revenue of online advertisement.



5- Browser Hijacker

- The browser hijacker malware can also be used for stealing personal information, user accounts, and other information for financial benefits.
- The symptoms and impact of browser hijack malware may include:

=Slow browsing speed

- Multiple tool bars on the browser
- Redirecting your search queries to websites that you have not set as default
- A large number of popup windows and ads appear on your browser

You can **save your browser from being hijacked** by the browser hijacker

malware by taking the following measures:

- Avoid free downloads as much as possible
- Disable unwanted tool bars
- Remove or disable unwanted plugins
- Set your default search engine
- Install antimalware erasers
- Use common sense while downloading and internet browsing

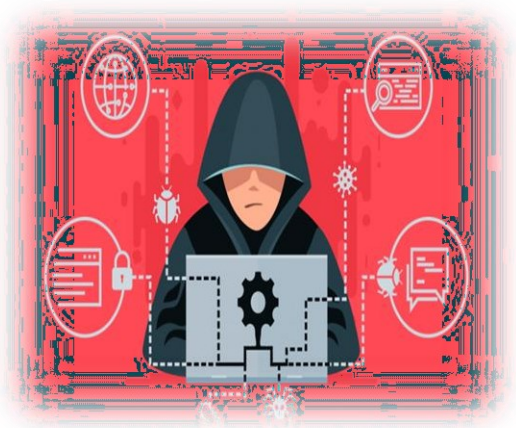




6- Rootkit

❖ There are several types of malware, the most important are as follows:

- **Rootkit**:- A class of malware that hides the existence of other malware by modifying the underlying operating system.
- **Rootkit** is a type of malware that gets the administrator level privileges on the operating system of the computer without showing its presence on the computer.
- The **main feature of rootkit** is that it hides from being detected easily but maintains the control over the operating system to perform its designated tasks on the system.
- The normal behavior of the operating system is subverted by the rootkit malware on the system.



6- Rootkit

- Moreover, you request the operating system to start the antivirus or anti- malware application to scan your computer; the request is interrupted from reaching to the operating system. The rootkit interrupts the request and sends you a fake response that the program you requested for is not working at this time.
- Rootkit is considered as one of the nastiest forms of malware that is not easily detected and removed from your computer once it has established the privileges to access and control the operating system through available vulnerabilities in the OS and other applications.

There are **three main goals of a rootkit** on a computer as mentioned below:

- Running freely without any restrictions of having been caught and deleted.
- Hiding from the system applications and the user of the computer
- Stealing personal information, passwords, and installing other malicious programs on the computer so that the compromised computer can be used for attacking the other computers on the network.



6- Rootkit

- There are numerous types of rootkits based on the objective and point of attack on the systems.
- A few **very well-known and important rootkits** are listed below:
 - Memory rootkits , Kernel rootkits , Bootkit rootkits , Firmware rootkits , Library rootkits Application rootkits
- For a **normal user** of the computer, it may be very difficult to recognize that the computer has been infected by the rootkits.
- If the computer shows the following **symptoms**, then you should suspect that the computer has been infected by the rootkit attack.
 - Modifications in the dates and time of computer
 - Slowdown of the computer
 - Unexpected system error messages
 - Many programs, especially computer security related programs fail to start
 - Substantial redirects on browsers
 - Appearance of blue screen of death





6- Rootkit

- It is very difficult to detect and remove the rootkit from your computer, once it has established the control over the operating system APIs.
- Normally, the rootkit remains unnoticed on the machine and keeps doing its job.
- For remaining many days on the computer, the system becomes a rogue machine, and many malicious programs start running on the machine. Those programs will be used for different malicious activities.
- You will find the behavior of operating system very irritating with the passage of time. One day you will see a cryptic blue screen of death and your computer will never load again normal.
- Now, you will have to re-install and configure your computer from the scratch.

- Many new software tools have also been created by the major computer security software providers like Norton, Avast, Panda, Kaspersky, and others.
- All those tools are able to detect the rootkits if present on your computer. They are also able to remove the nastiest forms of the rootkits on your computer easily.

7- Scareware

- **Scareware** is a type of malware, which pops up in window with a serious warning about any virus threat on your computer.
- **But**, in reality there is no threat or virus on your computer excepting that **hoax**, which appeared on your screen.
- The **alert looks** very genuine from certain reputable websites, but they are not genuine websites.
- This message normally prompts the users to download or call some numbers to get help.
- The **main objective behind the Scareware** is sale fake and bogus products.
- The use of scareware has been also found in the entertainment pranks.



7- Scareware

❑ Use of the Scareware:

(1) The hackers trick the users to input the credit card, personal, and bank information on their website. Once you provide that information, your data has been compromised for malicious financial uses.

(2) To force the user to download some free antivirus software to clean your computer from those viruses that the website has detected. But, that free antivirus is itself a dangerous software program, which can control your computer and start damaging your system or stealing your data.

- So, the **scareware** can be the foundation of many dangerous cyberattacks.

7- Scareware

The main objectives of spreading scare may include the following

- Selling fake products
- Stealing personal and bank information
- Installing viruses and other malicious codes for cyber crimes
- Blackmailing for some ransom
- Prank for fun



- The hackers use the social engineering tactics to force you follow the instructions they give you.
- So, always use your common sense and remain calm and cool to deal with the situation.

Other Attacks Types

- In addition to malware, there are many other types of attacks. The MITRE Corporation publishes a catalogue of attack patterns known as Common Attack Pattern Enumeration and Classification (CAPEC) as “an abstraction mechanism for helping describe how an attack against vulnerable systems or networks is executed.”
- Some of the most common attack patterns are listed below.

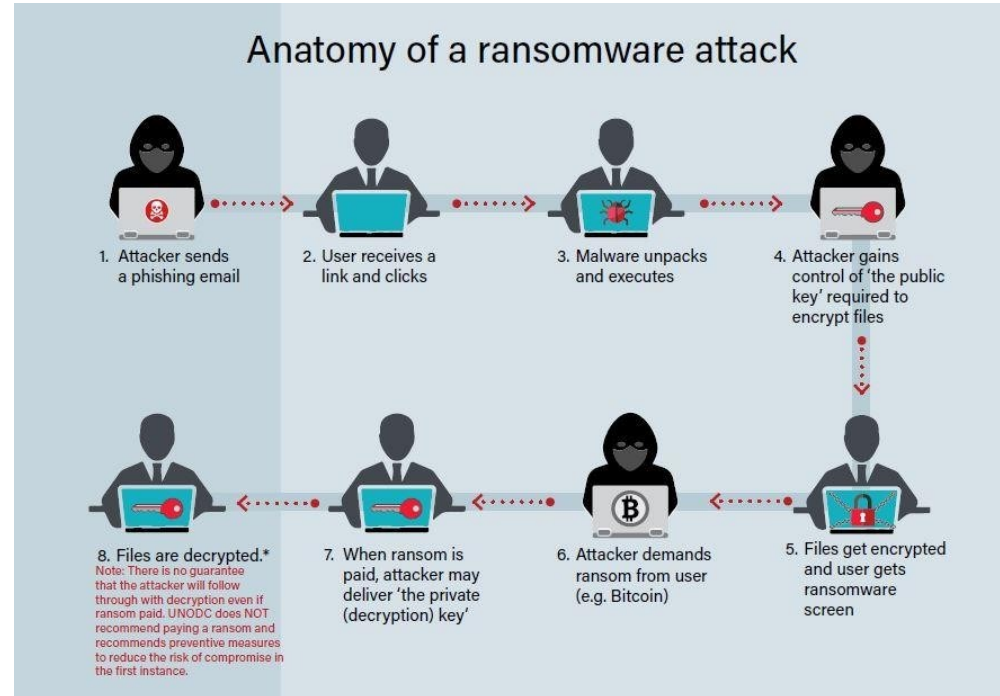


8- Ransomware

❖ There are several types of malware, the most important are as follows:

➤ A number of further terms are also used to describe more specific types of malware, characterized by their purposes. They include:

- **Ransomware:-** A class of extortive malware that locks or encrypts data or functions and demands a payment to unlock them.
- operating system.



9- Keylogger

- **Keylogger:-** A class of malware that secretly records user keystrokes and, in some cases, screen content.
- A recent example of malware's ability to function as a tool for cyberespionage is Flame, also known as Flamer and Skywiper. Discovered in 2012, it can record keyboard activity and network traffic as well as screenshots, audio and video communications such as Skype.
- Once collected, the recorded information is sent to various control servers, and a “kill command” is launched to wipe all traces of the malware from the computer.



10- Backdoor

- **Advanced persistent threats:-** Complex and coordinated attacks directed at a specific entity or organization. They require an enormous amount of research and time, often taking months or even years to fully execute.
- **Backdoor:-** A means of regaining access to a compromised system by installing software or configuring existing software to enable remote access under attacker-defined conditions.



Brute Force Attack

- **Brute Force Attack:-** An attack made by trying all possible combinations of passwords or encryption keys until the correct one is found.
- **Buffer Overflow:-** Occurs when a program or process tries to store more data in a buffer (temporary data storage area) than it was intended to hold. Since buffers are created to contain a finite amount of data, the extra information—which has to go somewhere—can overflow into adjacent buffers, corrupting or overwriting the valid data held in them. [In buffer overflow attacks, the extra data may contain codes type of security attack on data integrity.](#)

HOW A BASIC BRUTE FORCE ATTACK WORKS



Attacker



Guess List Of
Username &
Password
Combination



Repeats Login
Attempts Until
One Is Successful

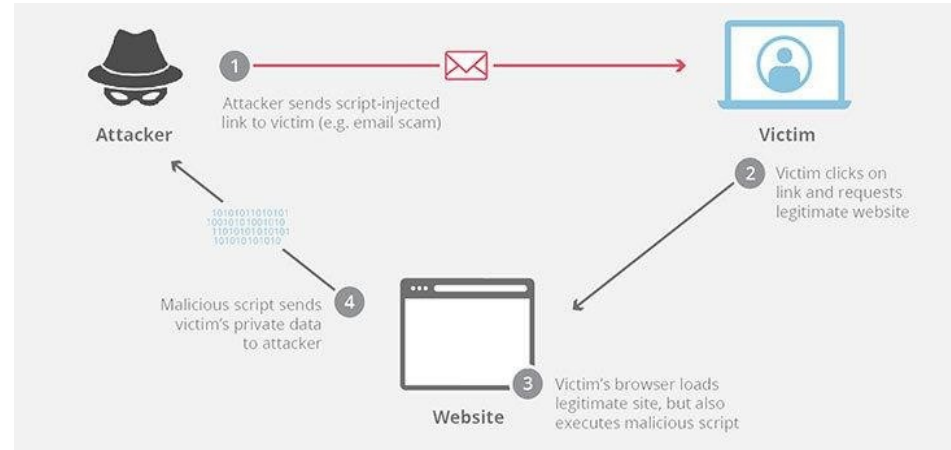


Successful
Credential Validation



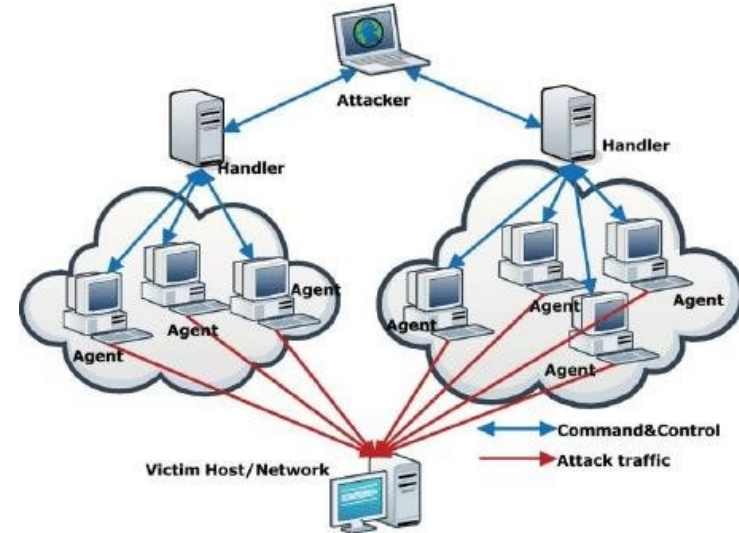
Cross-site scripting (XSS)

- **Cross-site scripting (XSS):-** A type of injection in which malicious scripts are injected into otherwise benign and trusted web sites. XSS attacks occur when an attacker uses a web application to send malicious code, generally in the form of a browser side script, to a different end user. Flaws that allow these attacks to succeed are quite widespread and occur anywhere a web application uses input from a user within the output it generates without validating or encoding it.



Denial-of-service (DoS) attack

- **Denial-of-service (DoS) attack:-** An assault on a service from a single source that floods it with so many requests that it becomes overwhelmed and is either stopped completely or operates at a significantly reduced rate.
- **Denial of service** or **DoS** is an internet security related event in which the hackers attack a particular server running some internet services to prevent it from working normal or stop the services.
- In this case the servers are overwhelmed with the flooding of superfluous messages.
- The hacker exploits the server vulnerability and sends the bombardment of automated requests and messages to that particular server to respond.



Denial-of-service (DoS) attack

- In **certain cases**, the service stops working any more due to overloaded server.
- These kinds of malicious attacks prevent the legitimate users from accessing the online services by either disrupting the services or stopping the services completely.
- The **major symptoms** of having become the victim of DoS attacks include the following:
 - Inability in accessing a website
 - Delay in accessing online service
 - Huge delays in file opening on the websites
 - Increased volume of spam emails
 - Degradation of performance of services

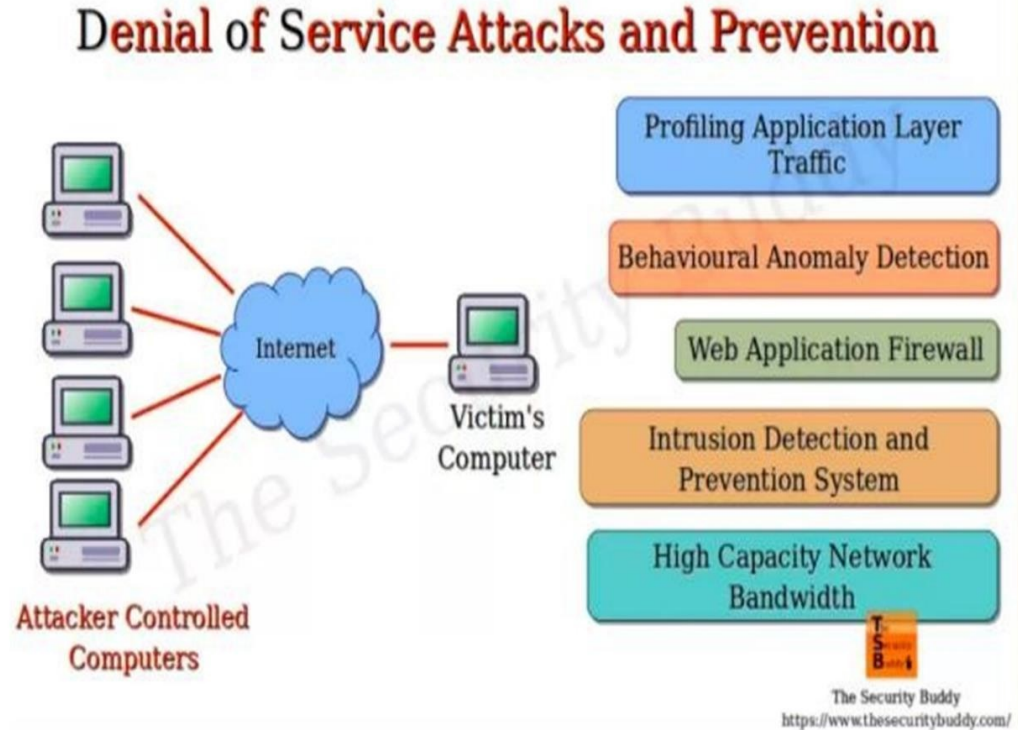
Signs of a DoS attack



Denial-of-service (DoS) attack

The impact of the DoS attack can be *mitigated* by taking the following steps:

- Routing the malicious traffic.
- Using load balancers to avoid heavy malicious traffic.
- Using intrusion detection systems (IDS)
- Using intrusion prevention systems (IPS)
- Using security firewalls



Denial-of-service (DoS) attack

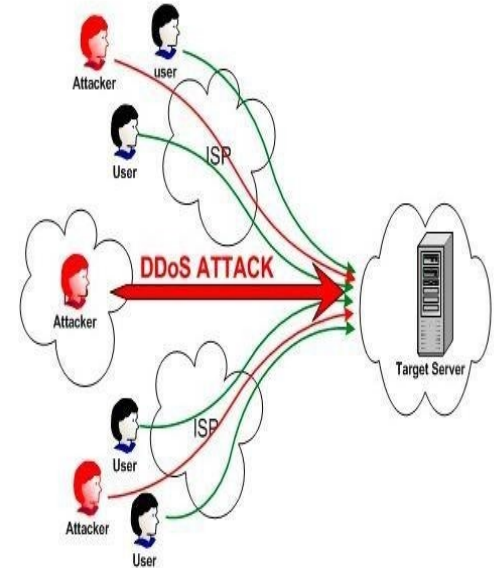
Main types of DoS attacks include the following:

- DNS server attack
- HTTP server attack
- ICMP protocol flooding
- Network attack or buffer overflow attack
- Large name files attack on the network or server
- Ping of death attack
- SYN flood attack on TCP handshake protocol



Distributed Denial of Services (DDoS)

- Distributed Denial of Service or precisely DDoS is also a type of DoS attacks.
- In which the servers are jammed or overwhelmed with the malicious traffic to prevent the legitimate users from accessing their accounts or legitimate online services.
- **Main different between DoS and DDoS attacks** is that the DoS attack is targeted from a specific origination of traffic to attack the victim server, while in the DDoS attacks multiple sources of traffic are used to attack the victim server.
- DDoS is more lethal than the DoS attacks. The prevention of DDoS attacks is very difficult as compared to the normal DoS attacks.



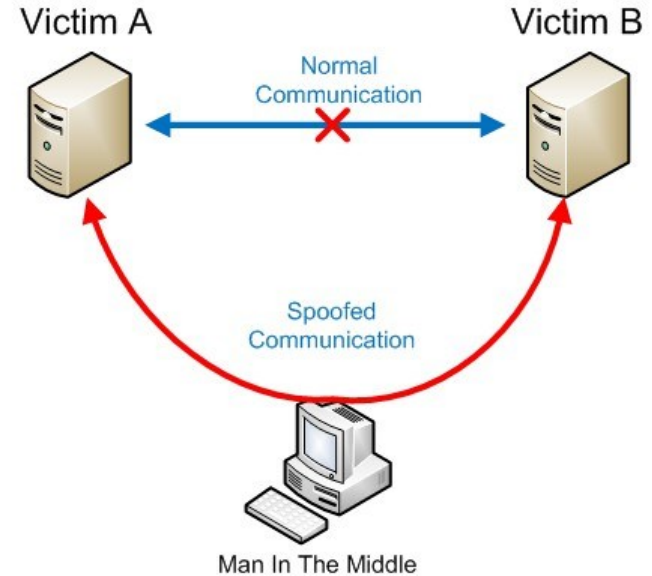
- In the distributed DoS attack, the hacker uses **controller machine** to hack multiple vulnerable machines around the globe.
- Then all those infected machines will be controlled by the software program running on the attacker's server.
- That controller sends instructions to the infected machine on which the hacker has already established the control to send automated requests (traffic) to the targeted victim machine.
- Then all those zombies start attacking the target from multiple IP sources to bring the server to almost halt.
- In this attack many computers from different parts become the part of this deadly attack without any approval and knowledge of the owner of the computer.

Man-in-the-middle attack

- **Man-in-the-middle attack:-** An attack strategy in which the attacker intercepts the communication stream between two parts of the victim system and then replaces the traffic between the two components with the intruder's own, eventually assuming control of the communication.

MITM attack consists of three major steps.

1. The hacker scans the vulnerabilities of the system and network.
2. Once the vulnerabilities are known, the phishing emails are sent to users; those emails contain the wrong addresses of your services or bank accounts.
3. Decrypt your information for stealing purpose.



Example MITM attacks:

Sometimes, you get an email that looks like it was sent from your bank or your other financial institute.

That is phishing email, which asks you to click certain link for the verification of your account, phone number or any other information.

Once you click that link, the link takes you to the web server, which looks like your bank website but in real, that is not your bank website.

You insert your credentials to login.

Thus, you provide your bank information to the hacker.

Finally, the hacker decodes your password and uses that password to take some from your account.

This entire process is known as the man in the middle or MITM cyberattack.

Man-in-the-middle attack

The major types of MITM attacks

- ❖ DNS Spoofing
- ❖ HTTP Spoofing
- ❖ IP Spoofing
- ❖ Email Hijacking
- ❖ SSL Hijacking
- ❖ Wi-Fi network Eavesdropping
- ❖ Stealing the cookies set on the browsers



Social engineering

- **Social engineering:-** Any attempt to exploit social vulnerabilities to gain access to information and/or systems. It involves a “con game” that tricks others into divulging information or opening malicious software or programs.



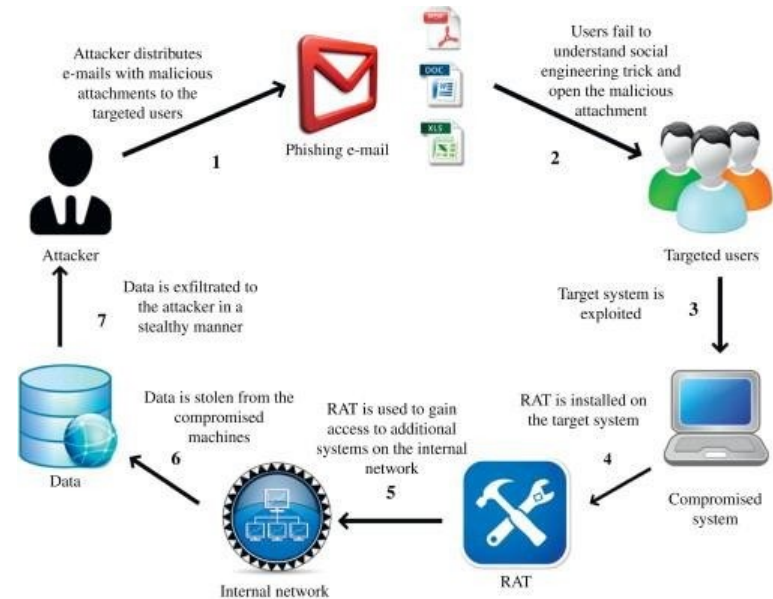
Phishing

- **Phishing:-** A type of electronic mail (email) attack that attempts to convince a user that the originator is genuine, but with the intention of obtaining information for use in social engineering.
- **The main target of the phishing attack** is to get the information about the credit card number, ATM pin codes, passwords, username, and the related information.



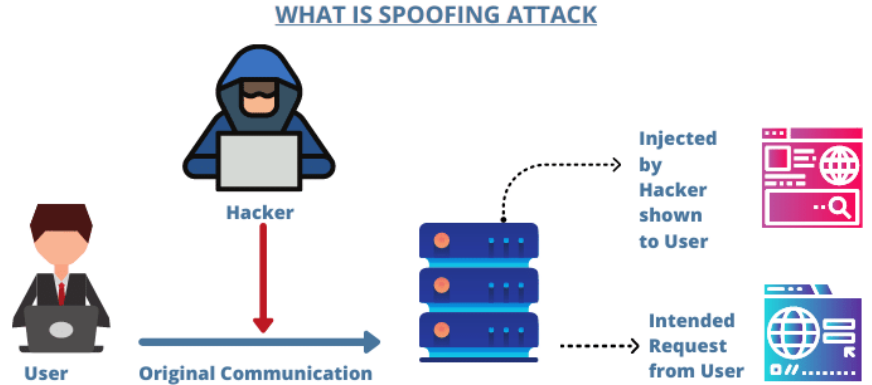
Spear phishing

- **Spear phishing:-** An attack where social engineering techniques are used to masquerade as a trusted party to obtain important information such as passwords from the victim.



Spoofing

- **Spoofing:-** Faking the sending address of a transmission in order to gain illegal entry into a secure system.

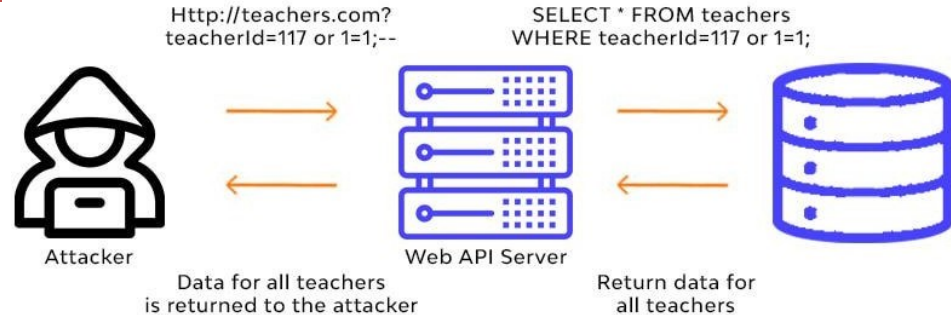


Structure Query language(SQL)injection

- **Structure Query language (SQL) injection:-**

According to MITRE, SQL injection results from failure of the application to appropriately validate input. When specially crafted user-controlled input consisting of SQL syntax is used without proper validation as part of SQL queries, it is possible to glean information from the database in ways not envisaged during application design.

SQL Injection



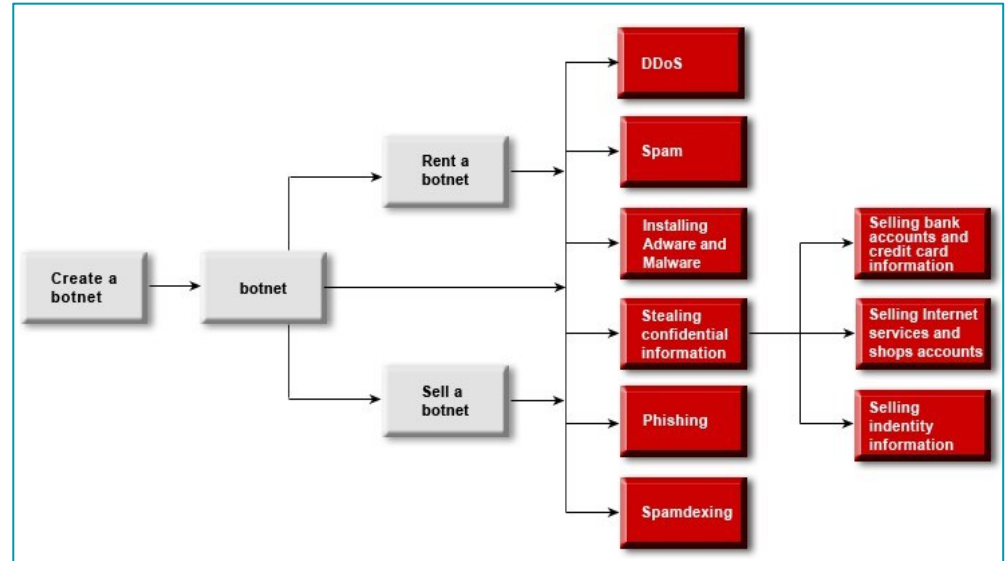
Zero-day exploit

- **Zero-day exploit:-** A vulnerability that is exploited before the software creator/vendor is even aware of its existence.
- Zero Day Exploitation commonly known as Zero-Day or 0-Day is a vulnerability in the computer software system that is known exactly on the same day when the malicious attacks exploit that vulnerability.
- In this attack, there is almost no time to patch up the vulnerability of the software because it was known at the same time when the attack occurred, and no time was available for the software engineers to tackle this issue.



Botnets

- **Botnets:-** A botnet (a term derived from “robot network”) is a large, automated and distributed network of previously compromised computers that can be simultaneously controlled to launch large-scale attacks such as denial-of-service (DoS).



Vulnerabilities

Vulnerabilities are the gaps or weaknesses in a system that make threats possible and tempt threat actors to exploit them.

Types of vulnerabilities in network security include but are not limited to

- SQL injections,
- server misconfigurations,
- cross-site scripting,
- and transmitting sensitive data in a non-encrypted plain text format.

Some of the most common network vulnerabilities include the following gaps in your application security: when applications are not kept up-to-date, tested, and patched, the doors are open to code injection, cross-site scripting, insecure direct object references, and much more.



Assignment

- Crypto-jacking
- Spamming
- Digital Property Misappropriation
- Digital Vandalism
- Cyberstalking
- Cyber Frauds & Forgery



Quiz, ex.

1. In, the digitally stored documents are forged to form the counterfeit documents. This crime is increased during the recent years due to the availability of high-tech devices like computer software, printers, scanners, cameras, and other tools.
 - A. cyber frauds and forgery
 - B. digital property misappropriation
 - C. zero-day exploitation
 - D. spamming

2. The impact ofin Asian countries is very high due to the traditional social structures.
 - A. cyber terrorism
 - B. digital property misappropriation
 - C. cyberstalking
 - D. Spamming



3. Cyberstalking is a systematic approach ofdone through email, phones, SMS, chats, and other forms of communication. The main components used in the communication to threaten the targeted entity include defaming, false allegations, slandering, and other forms of blackmailing.

- A. cyber terrorism
- B. digital property misappropriation
- C. zero-day exploitation
- D. harassing

4. In the attack the malware either remove the useful data from the websites or manipulate the information in such a way that the meaning of the information is reversed.

- A. cyber terrorism
- B. digital property misappropriation
- C. zero-day exploitation
- D. digital vandalism

The End 😊